

Conjectures on Prime Representation by Polynomials

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Postulate Statement

Postulate 1.1: Simple Polynomial Representation of Primes

For each fixed degree $d \geq 1$, there exists a bound $B(d)$ such that every prime $p \geq B(d)$ can be expressed as

$$p = a_d x^d + a_{d-1} x^{d-1} + \cdots + a_1 x + a_0$$

for some natural numbers $a_0, a_1, \dots, a_d$ and x , with $\gcd(a_0, a_1, \dots, a_d) = 1$.

Note: This postulate is considered trivial in the sense that it can be readily established constructively. For any prime p and degree d , we can always find coefficients and evaluation point to represent p . The postulate serves as a foundation for the more substantive conjectures that follow.

Conjecture Statements

Conjecture 1.2: Strong Form with Constraint Sets

Formal Statement:

For each fixed degree $d \geq 1$, let S be any set of conditions on $(d+2)$ -tuples $(a_0, a_1, \dots, a_d, x)$ where the a_i are natural numbers with $\gcd(a_0, a_1, \dots, a_d) = 1$.

If there exist infinitely many tuples satisfying S , then there exists a bound $B(d, S)$ such that for every prime $p \geq B(d, S)$, the number of valid tuples $(a_0, a_1, \dots, a_d, x) \in S$ satisfying

$$p = a_d x^d + a_{d-1} x^{d-1} + \cdots + a_1 x + a_0$$

grows as p increases beyond $B(d, S)$.

Formal Notation:

$\forall d \in \mathbb{N}^+, \forall S \subseteq \mathbb{N}^{d+2}$ with $|S| = \infty$ and gcd condition,

$\exists B(d, S) \in \mathbb{N} : \forall p_1, p_2$ prime with $B(d, S) \leq p_1 < p_2$,

$$|\{(a_0, \dots, a_d, x) \in S : p_1 = \sum_{i=0}^d a_i x^i\}| < |\{(a_0, \dots, a_d, x) \in S : p_2 = \sum_{i=0}^d a_i x^i\}|$$

Conjecture 1.3: Refined Form with Signed Coefficients

Formal Statement:

For each fixed degree $d \geq 1$, let S be any set of conditions on $(d + 2)$ -tuples $(a_0, a_1, \dots, a_d, x)$ where:

- a_0 and a_d are natural numbers
- The intermediate coefficients $a_1, \dots, a_{d-1}$ are integers (possibly negative)
- $\gcd(a_0, a_1, \dots, a_d) = 1$

If there exist infinitely many tuples satisfying S , then **for every prime p** (without any lower bound), there exist **infinitely many** tuples $(a_0, a_1, \dots, a_d, x) \in S$ satisfying

$$p = a_d x^d + a_{d-1} x^{d-1} + \dots + a_1 x + a_0$$

Formal Notation:

$$\forall d \in \mathbb{N}^+, \forall S \subseteq (\mathbb{N} \times \mathbb{Z}^{d-1} \times \mathbb{N} \times \mathbb{N}) \text{ with } |S| = \infty,$$

$$\forall p \text{ prime,}$$

$$|\{(a_0, \dots, a_d, x) \in S : p = \sum_{i=0}^d a_i x^i\}| = \infty$$

Context and Related Work

These conjectures represent a novel perspective on the relationship between prime numbers and polynomials, approaching from an inverse direction compared to classical results.

Known Results:

1. **Dirichlet's Theorem (1837):** Linear polynomials $P(n) = an + b$ with $\gcd(a, b) = 1$ generate infinitely many primes.
2. **Bunyakovsky Conjecture (1857):** An irreducible polynomial with integer coefficients should generate infinitely many primes if no fixed prime divides all values.

3. **Jones-Sato-Wada-Wiens (1976):** There exists a polynomial in 26 variables whose positive values are exactly the set of all primes (Diophantine representation).
4. **Schinzel's Hypothesis H:** A generalization of Bunyakovsky's conjecture to systems of polynomials.

Distinction from Related Problems

Classical results ask: "**Does this polynomial generate primes?**"

These conjectures ask: "**Can primes be represented by polynomials under constraints?**"

The key differences:

- **Classical approach:** One polynomial $\rightarrow$ many primes
- **Proposed approach:** Many polynomials $\rightarrow$ all primes

Specific distinctions:

1. **Direction reversal:** Instead of asking whether a fixed polynomial hits infinitely many primes, we ask whether all primes get hit by some polynomial (infinitely many times).
 2. **Constraint flexibility:** We investigate whether constraint sets on polynomial coefficients are "rich enough" to maintain universal prime coverage.
 3. **Abundance of representations:** We don't just ask for existence but assert that the number of representations grows with the prime's magnitude.
 4. **Degree-dependent bounds:** Recognition that the bound $B(d)$ or $B(d, S)$ naturally depends on polynomial degree.
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Significance

If True:

These conjectures would establish that:

1. **Polynomial expressiveness:** The space of polynomials with coprime coefficients (even under constraints) is sufficiently rich to represent all primes.
2. **Flexibility principle:** If a constraint set S doesn't reduce solutions to a finite set, it maintains enough degrees of freedom such that larger primes admit more polynomial representations.
3. **Universality of signed representations:** Allowing signed intermediate coefficients eliminates all bounds, enabling representation of every prime.
4. **Density insights:** There exists a fundamental abundance in how primes can be expressed polynomially.

If False:

A counterexample would reveal:

1. **Fundamental limitations:** Certain constraint sets, even if infinite, cannot reach all primes.
2. **Prime-polynomial barriers:** There exist structural obstructions preventing universal polynomial representation.
3. **Degree limitations:** For some degrees, the polynomial space is insufficiently expressive.

Relationship Between Conjectures

The three statements form a natural hierarchy:

$$\text{Conjecture 1.3} \implies \text{Conjecture 1.2} \implies \text{Postulate 1.1}$$

Each successive statement strengthens the previous:

- **Postulate 1.1:** Establishes basic representability for large primes (single representation suffices)
- **Conjecture 1.2:** Adds constraint robustness and shows representations grow with prime size
- **Conjecture 1.3:** Removes all prime-size bounds via signed coefficients and guarantees infinitely many representations

Key progression:

Statement	Coefficients	Bound on p	Representations
Postulate 1.1	All natural	$p \geq B(d)$	At least one
Conjecture 1.2	All natural	$p \geq B(d, S)$	Growing with p
Conjecture 1.3	Signed (except a_0, a_d)	None	Infinitely many

Open Questions

1. **Computational bounds:** Can we establish explicit formulas for $B(d)$ or $B(d, S)$ in terms of degree and constraint complexity?
2. **Degree progression:** What is the minimum degree d for which Postulate 1.1 holds with $B(d) = 0$ (all primes representable)?
3. **Constraint characterization:** Which constraint sets S are "too restrictive" (have infinite solutions but cannot represent certain primes)?
4. **Algorithmic construction:** Is there an effective algorithm to construct a polynomial representation for a given prime p and constraint set S ?
5. **Density formulas:** Can we derive asymptotic estimates for the number of distinct polynomial representations of a prime p as $p \rightarrow \infty$?

6. **Relationship to classical conjectures:** How do these conjectures relate to Bunyakovsky, Schinzel's Hypothesis H, and the Hardy-Littlewood conjectures?
 7. **Extension to composite numbers:** Do similar results hold for composite numbers, or do primes have special polynomial representation properties?
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Examples and Special Cases

Linear Case ($d = 1$):

For $d = 1$, we have polynomials $P(x) = ax + b$.

- **Postulate 1.1:** Every prime p can be written as $ax + b$ for some coprime (a, b) and x .
 - Trivially true: $p = 1 \cdot 1 + (p - 1)$ or $p = 1 \cdot 0 + p$.
- **Conjecture 1.2:** For constraint sets with infinitely many solutions, every large prime has infinitely many linear representations.
 - Related to Dirichlet's theorem but asks the inverse question.

Quadratic Case ($d = 2$):

For $d = 2$, polynomials $P(x) = ax^2 + bx + c$.

This case is particularly interesting as it relates to classical questions about primes of the form $ax^2 + bx + c$ (e.g., primes of the form $n^2 + 1$).

Methodology and Approach

Potential approaches to these conjectures:

1. **Constructive methods:** Develop explicit algorithms to construct polynomial representations.
 2. **Density arguments:** Use analytic number theory to show that the space of valid $(a_0, \dots, a_d, x)$ tuples is dense enough.
 3. **Sieve methods:** Employ sieve theory to control both prime and composite generation.
 4. **Combinatorial arguments:** Count the degrees of freedom in coefficient space versus constraints.
 5. **Connection to existing conjectures:** Explore whether proving/disproving these conjectures would impact Bunyakovsky or Schinzel's Hypothesis H.
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Status

OPEN - As of January 2026, these conjectures appear to be original and have not been addressed in this form in the

mathematical literature.

The inverse perspective (primes being represented by polynomials, rather than polynomials generating primes) represents a novel approach to understanding prime-polynomial relationships.

Acknowledgments

These conjectures arose from exploring the flexibility and expressiveness of polynomial representations in number theory, motivated by the observation that the space of polynomials with natural coefficients appears "large enough" to capture primes in multiple ways.

References

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